

Sixth-Sense Media Player



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Sixth sense or gestural interface program is a popular term especially in robotics and wearable computers. This project describes a hand-gesture-controlled media player. Many hand-gesture-controlled media players have been developed based on image processing techniques. However, in such a media player, the user moves the hand in front of the camera. Here, we present another technique to control the media player using a hand glove (Fig. 1). It



Fig. 1: Sixth-sense media player being operated using a hand glove

requires Windows Media Player, LabVIEW, Arduino and an accelerometer.

LabVIEW

Laboratory Virtual Instrument Engineering Workbench, popularly known as LabVIEW, is a system design platform and development environment for a visual programming language from National Instruments (NI). LabVIEW programs are called virtual instruments, or VIs.

LabVIEW works on the principle of data flow. It is basically a graphical programming language in which the user can set up the program to manipulate and store data. There are three important components in LabVIEW programming: Front Panel, Block Diagram, and Icons and Connectors as shown in Figs 2, 3 and 4, respectively.

Front Panel is used to simulate the process or virtual testing area. It serves as the user interface (UI). Here, indicators and controllers are shown without any connection between these. We can control the action or simulate the process by clicking Run on the Tools bar.

Block Diagram is used to program the process for the purpose of getting required or desired output. It contains the graphical source code that defines the functionality of the VI. Here, the programming method is graphical, so it is easy to understand, program and is less time-consuming.

Icons are used as variables or controllers. These play a vital role in programming. These icons have pre-defined functions. Wires in LabVIEW are used to connect the input to the output.

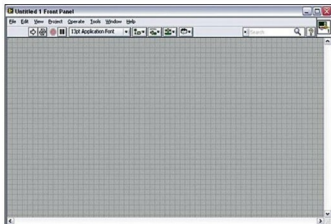


Fig. 2: Front Panel of LabVIEW

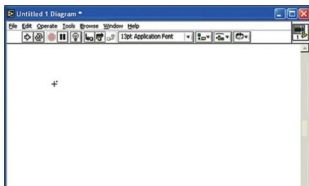


Fig. 3: Block Diagram of LabVIEW

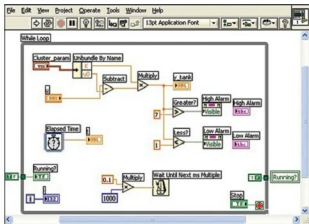


Fig. 4: Icons and Connectors of LabVIEW